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INPUT DATASET

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measurements.xlsx

sheet measurement_01

Variables

f

frequency

x-axis

Hz

U

voltage

input

V

I

current

input

mA

z

impedance

calculated

Ω

YOUR REQUEST

Calculate impedance for each measurement and create a chart of impedance vs frequency.

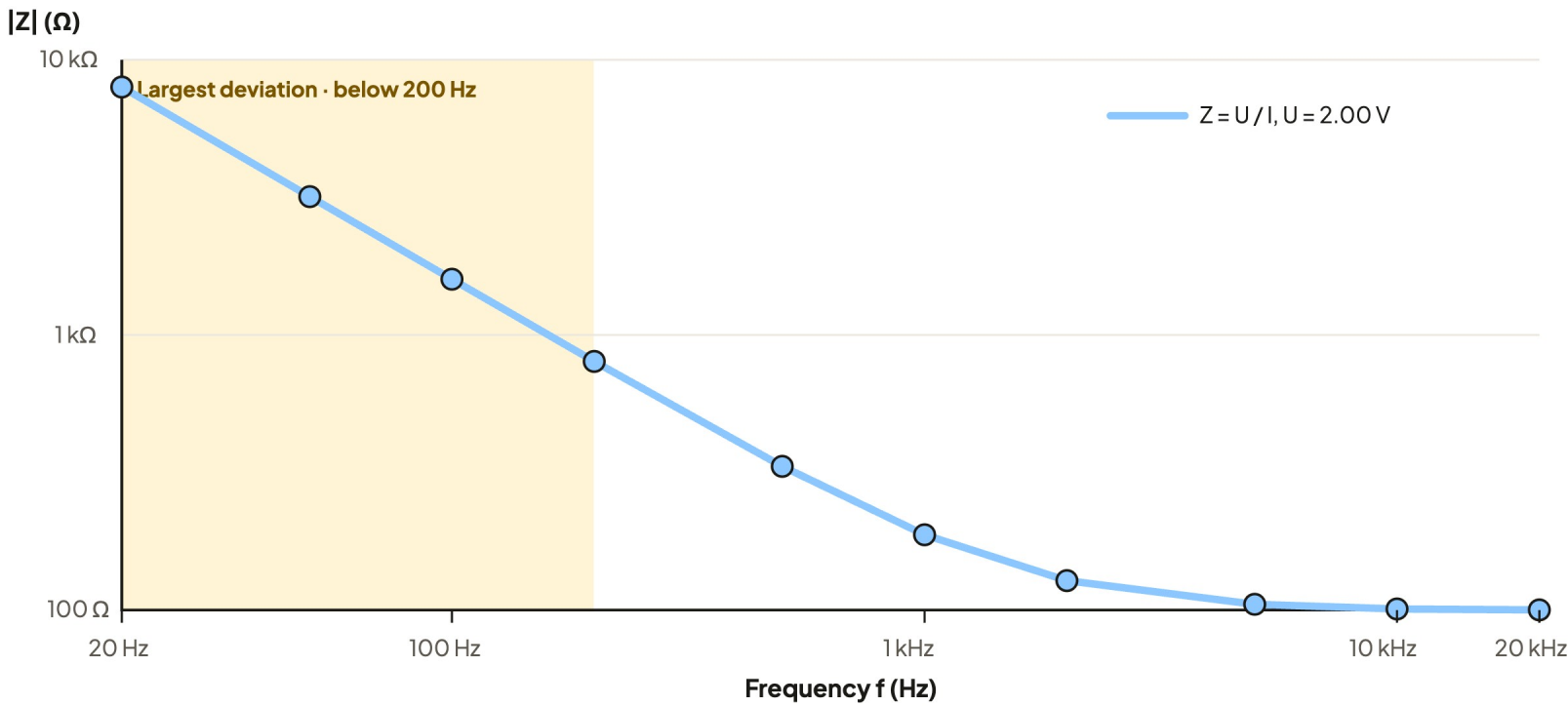
Analysis #17 ✓ Completed

Calculate impedance vs frequency

</> Hide code 🔄 Re-run 📄 Insert table 📈 Insert chart

Impedance magnitude |Z| versus frequency

log-log · 10 of 42 points marked



Result table

8 of 42 rows · sorted by frequency

frequency (Hz)	voltage (V)	current (mA)	impedance (Ω) · new
20	2.00	0.25	7958
50	2.00	0.63	3185
100	2.00	1.25	1595
200	2.00	2.49	802
500	2.00	5.99	334
1000	2.00	10.64	188
5000	2.00	19.07	105
20000	2.00	19.94	100

The code that produced this

Python · read-only

This is exactly what ran. Edit it to change the analysis, or re-run it to reproduce the result.

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 df = pd.read_excel("measurements.xlsx", sheet_name="measurement_01")
5 df = df[["frequency", "voltage", "current"]].dropna()
6
7 # Z = U / I (current is in mA, so convert to A first)
8 df["impedance"] = df["voltage"] / (df["current"] / 1000)
9
10 result = df.sort_values("frequency")
11 ax = result.plot(x="frequency", y="impedance", logx=True, logy=True, marker="o")
12 ax.set_xlabel("Frequency (Hz)"); ax.set_ylabel("|Z| (Ω)")
```

Data unchanged since this ran. If measurements.xlsx is replaced, this result is marked out of date.

Provenance

Source measurements.xlsx

Sheet measurement_01

Columns frequency, voltage, current

Operation $Z = U / I$

Parameters $U = 2.00 \text{ V}$ · log-log axes

Executed Today, 12:42 · 3.2 s

Created by Adam Nowak

Status Completed

Run history

- Re-run available

Same data, same code

now
- Completed

Adam · today 12:42

current
- Draft request

Adam · today 12:39

Used in

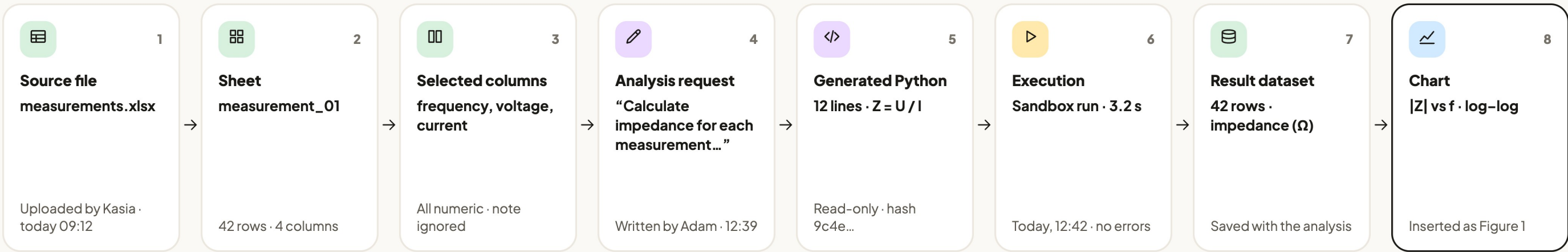
📄 Laboratory Report
4. Results · Figure 1

[See the full chain](#) from the source file to this chart.

Where this chart came from

Analysis #17 — every step from the source file to the figure, kept together and re-runnable.

Reproducible



Run history

RUN	WHEN · WHO	DATA VERSION	CODE	RESULT
Run 1	Today 12:42 · Adam	v3 · ab12...	9c4e...	Current
—	Today 12:39 · Adam	v3 · ab12...	draft	Replaced

Same data version and same code give the same result. If the data version changes, the chart in your document is marked **Out of date** until you refresh it.

Used downstream

- 📄

Laboratory Report
4. Results · Figure 1 · chart + caption

Up to date
- 📄

Draft Conclusions
Referenced in paragraph 2

Up to date

Re-run analysis

Export record

Below 200 Hz the impedance rises steeply, as expected for the capacitive branch of the circuit. Above 5 kHz it levels out near 100 Ω , the resistive limit. **A17**